

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

01 / PROBLEM

Why current agents fall short.

Sustained autonomous exploration encounters three critical bottlenecks:

01 Closed-world fragility

Rigid predefined pipelines shatter as soon as unexpected runtime errors emerge. In Cycle 1, a **float32 JSON serialization bug** halted logging; **without autonomous self-healing, exploration dies permanently**.

02 Single-trajectory inertia

Accumulating context history anchors the agent to past reasoning, **trapping search in local basins** (e.g., standard single-agent baselines **stall at 0.6044** while endlessly fine-tuning hyperparameters).

03 Context inflation cost

Continually rereading deep conversational context **costs 8x more tokens** than the actual code edits, exhausting context windows and causing **severe reasoning dilution**.

TRADITIONAL AGENT HARNESS

Single Monolithic Context · Error Cascades · Token Waste · Local Optima

SCIODYSSEY RESEARCH LAYER

Ephemeral Fresh Scientist · Persistent Disk Memory · Verified Handoff

PARADIGM SHIFT

Decouple the **persistent research world** (disk memory) from ephemeral **researcher reasoning** (fresh context).

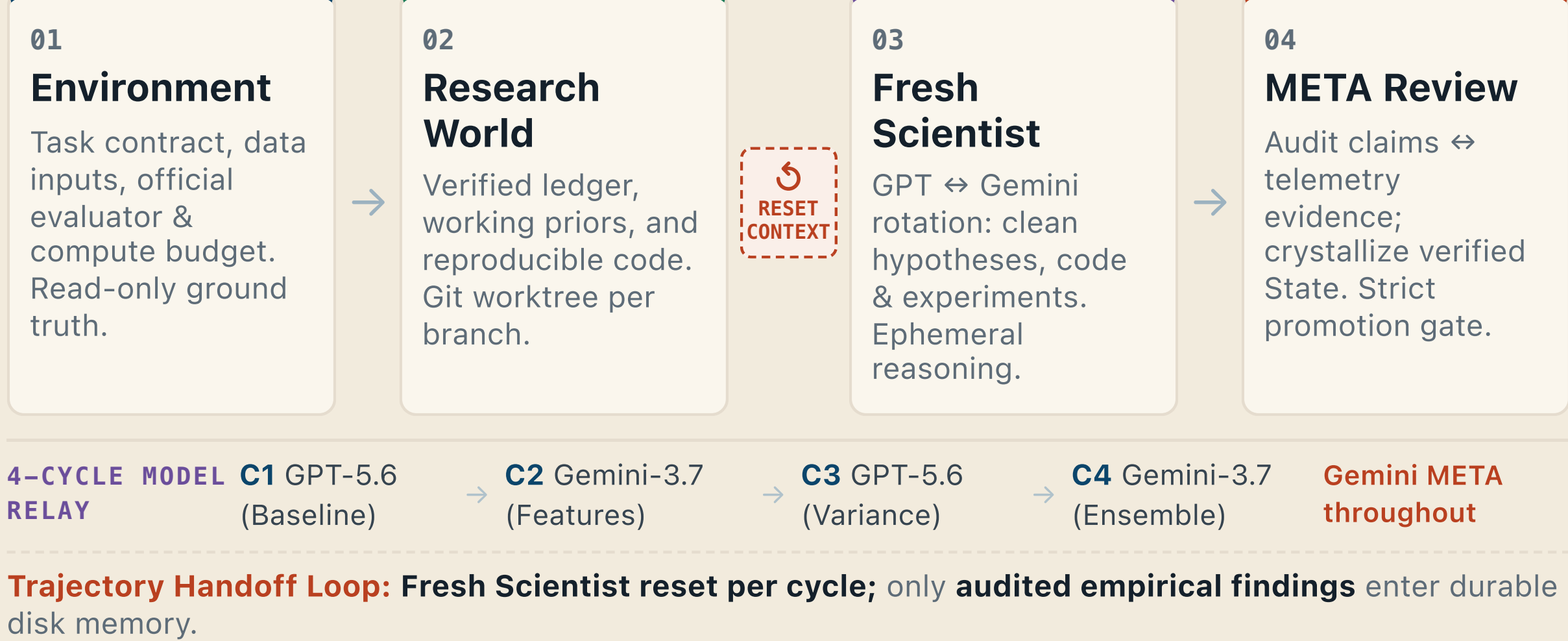
02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

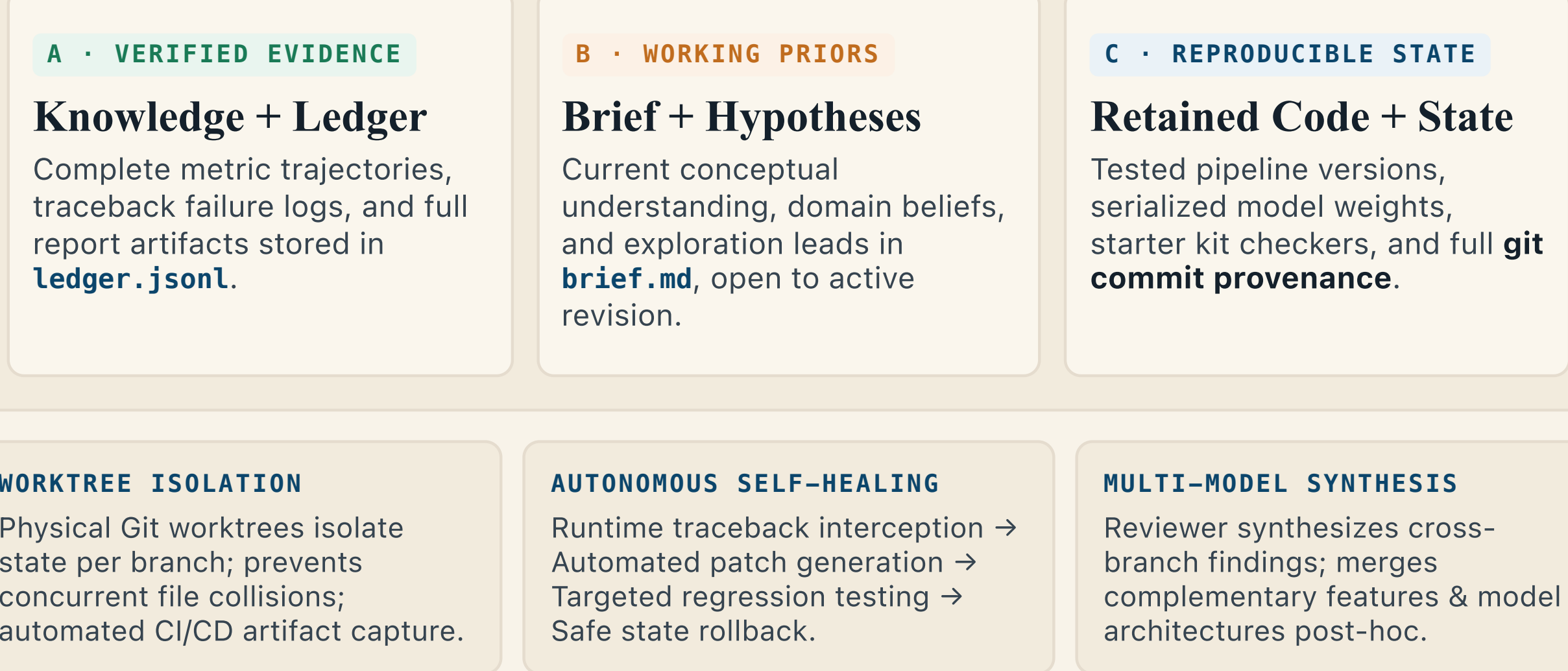
THE DUAL-LIFETIME ENGINE

Context reset between trajectories · Audited persistence on disk



PERSISTENT RESEARCH WORLD

What survives across trajectories



The next Scientist inherits **verified experience, not prior reasoning inertia**.
Decoupled Memory · Audited Persistence · Autonomous Self-Healing · Zero Context Pollution

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000

Official FM baseline



0.6059363

SciOdyssey submission

+0.0043363 Primary improvement

GAUC **0.6728421** (+0.0054) nDCG@5 **0.5390304** (+0.0033) LogLoss **0.38102** (-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms
Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)
Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); **100% CPU inference** without GPU clusters.

03 SELF-HEALING & GATEKEEPING

The agent recovered from failure.

TIER 1 · SCIENTIST SELF-HEALING Cycle 1 runtime
Traceback auto-repaired: numpy.float32 broke JSON serialization. Scientist diagnosed traceback, patched scalar serialization helper, and reran cleanly.

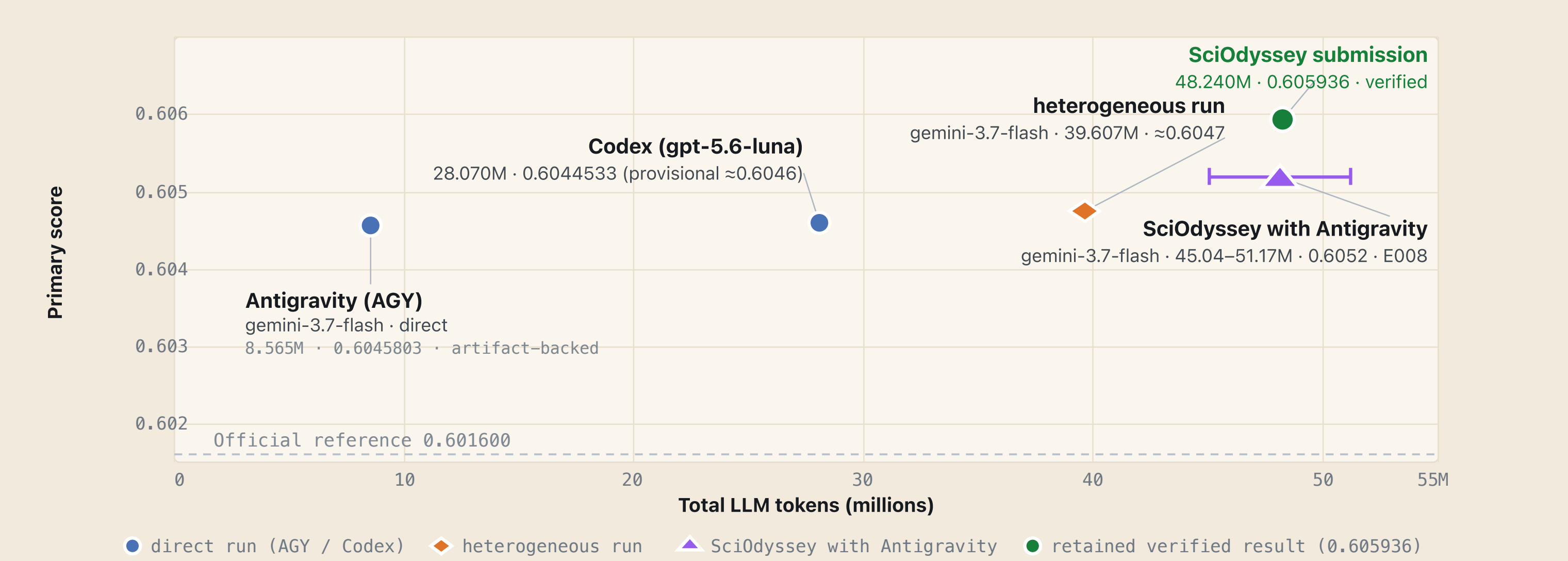
TIER 2 · META GATEKEEPING Proxy audit receipt
Spurious proxy rejected: Scientist claimed false gain; Meta audited system/data.py, caught UNK user sampling distortion, and blocked State promotion.

Self-Healing: 0 human bugfixes **META Gate:** Spurious proxy killed **Audit:** 170,588 rows aligned

Robust research requires both **execution-level crash recovery** and **epistemic defense** against false breakthroughs. **Zero human debugging interventions across 7 days.**

05 DIRECT AGENT BENCHMARK

Score versus measured LLM-token investment.



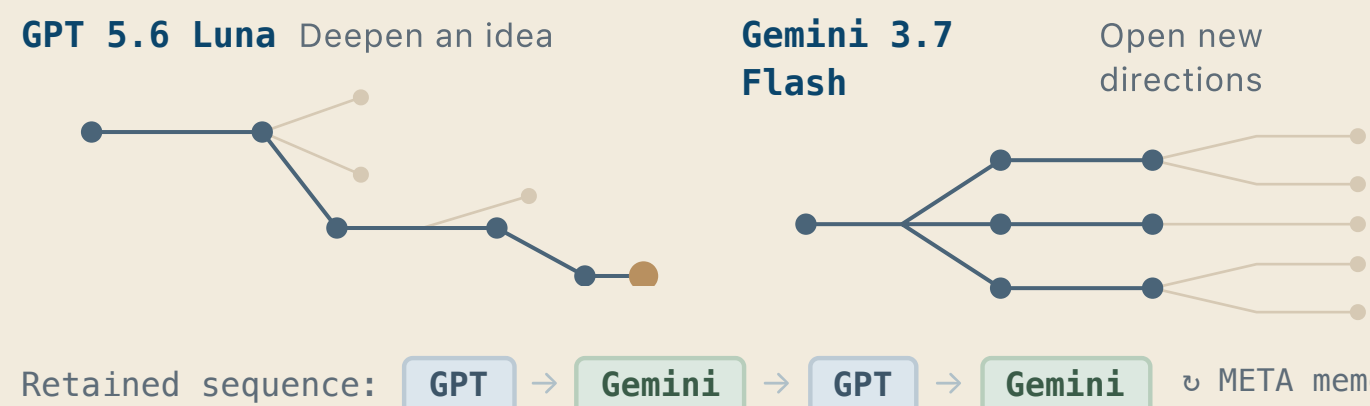
All results **verified on KuaiRand-Pure official validation split** with **complete reproducible artifacts**.

04 / INSIGHTS

What the run taught us.

01 · MODEL DIVERSITY

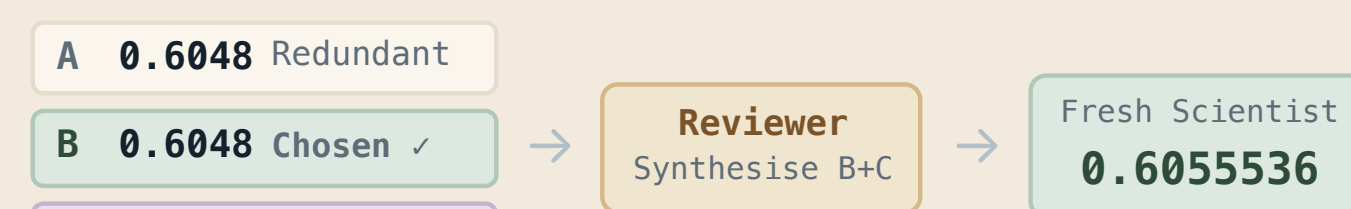
Depth + Breadth Synergy.



Gemini-only 0.6052 → GPT+Gemini 0.6059363

02 · PARALLEL SYNTHESIS

Cross-Branch Idea Transfer.



Lower-scoring Branch C contributed pairwise ranking and rank normalisation — a lower score can still carry a useful idea.

Branch B+C Ideas → Synthesis 0.6055536

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

- Dataset paths & split integrity
- Official sandbox evaluator

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

- Model rotation (GPT / Gemini)
- Supervision & budget thresholds

SUPERVISION TIERS

Tier 1 Unattended Overnight Runs · Tier 2 Checkpoint Review (META) · Tier 3 Steer Working Priors

1. Define task.md → 2. Single-step test → 3. Autonomous search → 4. Parallel explore → 5. Inspect evidence

DECLARATIVE PROTOCOL

Human defines boundary and preferences (task.md + PERSONAL.md); SciOdyssey autonomously iterates, heals, and crystallizes verified evidence.

Autonomous by default.

Supervisable by design.

Coding agent harness · Execution studio

Works with: Codex Claude Code Antigravity OpenCode Trae

> SKILL.md "Use the research-agent skill. Explore in parallel; let me review what to keep."

\$ research-agent step # one cycle · human reviews

META → Scientist → Evidence → Your turn ✓

\$ research-agent run --max-cycles 4 # autonomous · fresh scientist each round

C1 → C2 → C3 → C4 · NDCG 0.6059 ✓

\$ research-agent parallel --branches 3 # isolated worktrees · reviewer synthesises

Branch A-B-C → Reviewer Synthesis → +0.00148

\$ research-agent gui # dashboard · telemetry & checkpoints

Telemetry · Loss Curves · S004 Checkpoint ✓

Status: SOTA Verified (0.6059363) Harness: 100% Autonomous Hardware: 1x MacBook M2 (0 GPU-hr) Scope: 170,588 rows evaluated

